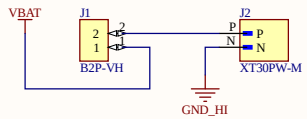
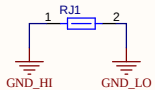
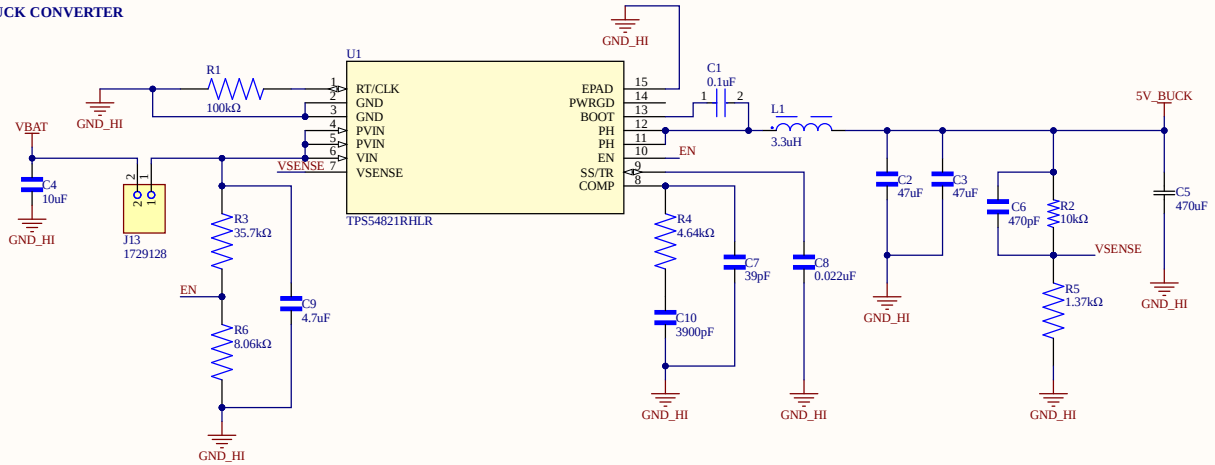
POWER SWITCH



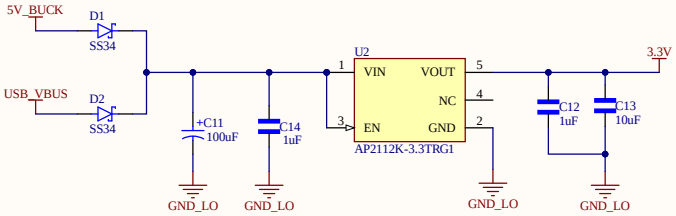
GROUND TIE



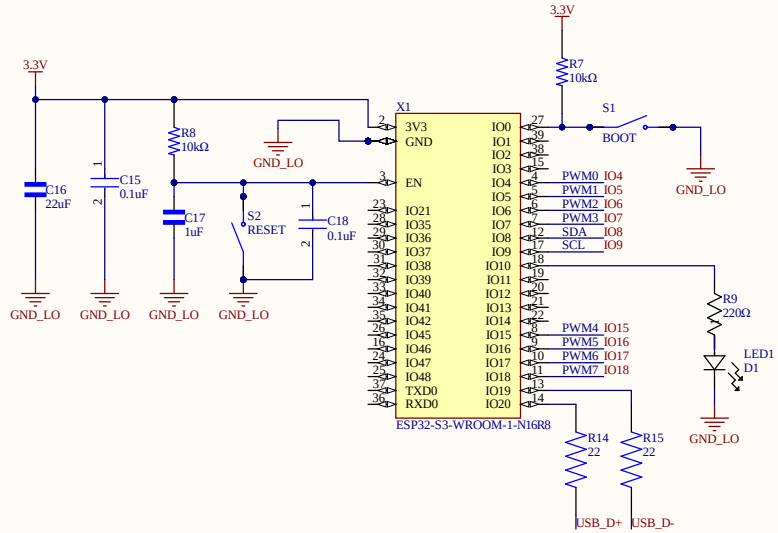
BUCK CONVERTER



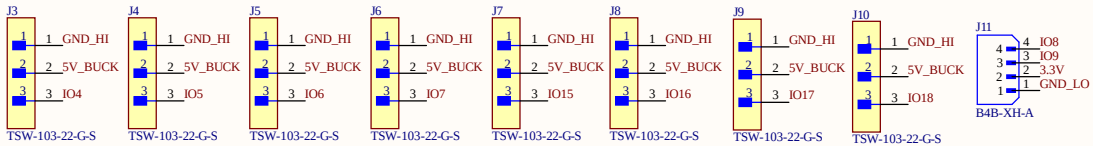
LOW-DROPOUT REGULATOR



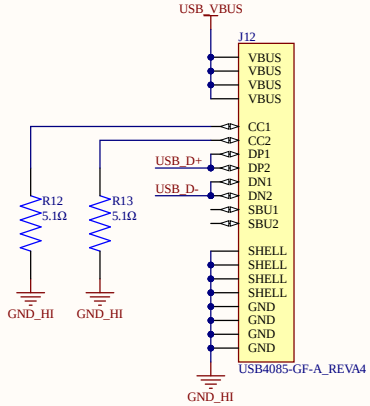
MICROCONTROLLER



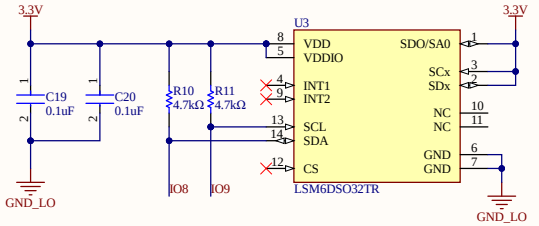
PERIPHERALS



USB-C CONNECTOR



IMU SENSOR



Title: Hermit_V1		
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Date: 6/04/2026		Sheet of
File: C:\Users\..\Hermit_V1.SchDoc		Drawn By: Emily Chan & Jesse Chen

Line#	Name	Description	Designator	Revision ID	Revision Status	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	GRM033R61A104KE15D	Chip Multilayer Ceramic Capacitors for General Purpose, 0201, 0.10uF, X5R, 15%, 10%, 10V	C1, C15, C18, C19, C20	CMP-2010-01264-2	Up to date	5	Murata	GRM033R61A104KE15D	Volume Production	Mouser	81-GRM033R61A104KE5D	0.01	0.1
	47uF	SMD capacitor (X7R/ElA) with capacitance 47uF Tol. 10%. Rated voltage 10Vdc 125C.	C2, C3			2							
	10uF	SMD Capacitor X5R(EIA) with Capacitance: 10uF Tol. 20%. Rated Voltage: 25Vdc	C4			1							
	4.7UuF	Description	C5			1							
	470pF	SMD Capacitor X7R(EIA) with Capacitance: 470pF Tol. 10%. Rated Voltage: 50Vdc	C6			1							
	10pF		C7			1							
	0.022uF	0.022 ÂµF Â±10% 16V Ceramic Capacitor X7R 0603 (1608 Metric)	C8			1							
	4.7uF	4.7 ÂµF Â±10%25V Ceramic Capacitor X5R0805 (2012 Metric)	C9			1							
	3900pF	3900 pF Â±1% 500V Ceramic Capacitor C0G, NP0 1206 (3216 Metric)	C10			1							
	100uF	SMCAFA91L Aluminum Electrolytic Capacitor, Radial, THT, 5mm x 11mm, 100uF, 10V	C11			1							
	1uF	1 ÂµF Â±10% 10V Ceramic Capacitor X5R0402 (1005 Metric)	C12, C14, C17			3							
	10uF	10 ÂµF Â±20% 10V Ceramic Capacitor X5R0402 (1005 Metric)	C13			1							
	22uF	SMD capacitor X5R(EIA) with capacitance 22uF Tol. 20%. Rated voltage 10Vdc 85C	C16			1							
	SS34	Diode Schottky 40 V 3A Surface Mount DO-214AB)	D1, D2			2							
	8SPVH	2 Position Adapter Connector Receptacle, Female Sockets Top Receptacle, Female Sockets 0.156 (3.96mm)	J1			1							
	XT60PW-M	Socket; DC supply; XT30; male; PIN: 2; on PCBs; THT; yellow; 15A; 500V	J2			1							
	TSW-103-22-G-S	Conn Unshrouded Header HDR 3 POS 2.54mm Solder ST Thru-Hole Bulk	J3, J4, J5, J6, J7, J8, J9, J10	CMP-02766-000544-1	Up to date	8	Samtec	TSW-103-22-G-S	Volume Production	Avnet	TSW-103-22-G-S	0.47922	3.83
	B4B-XH-A	Connector Header Through Hole 4 position 0.098 (2.50mm)	J11			1							
	USB4085-GF-A_REVA4	USB-C (USB TYPE-C) USB 2.0 Receptacle Connector 24 (16+8 Dummy) Position Through Hole, Right Angle, low-cost, 3.46mm profile, 16 µin, horizontal, Top Mount	J12			1							
	1729128	PCB Terminal Screw 5.08mm 16AWG 400V 13.5A 2 Position COMBICON MKDSN Series	J13			1							
	3.3uH	3.3 µH Shielded Molded Inductor 5.4 A 30mOhm Max 2-SMD	L1			1							
	D1	Description	LED1			1							
	100kΩ	100 kOhms Â±0.25% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Thin Film	R1			1							
	10kΩ	10 kOhms Â±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	R2, R7, R8			3							
	35.7kΩ	Res Thin Film 0603 35.7K Ohm 0.1% 0.1W(1/10W) Â±25ppm/°C Epoxy Pad SMD T/R	R3			1							
	4.64kΩ	Res Thin Film 0402 4.64K Ohm 0.1% 0.063W(1/16W) Â±25ppm/°C Molded SMD Paper T/R	R4			1							
	1.37kΩ	Res Thin Film 0402 1.37K Ohm 0.1% 0.063W(1/16W) Â±25ppm/°C Molded SMD Paper T/R	R5			1							
	8.06kΩ	8.06 kOhms Â±0.1%0.1W, 1/10W Chip Resistor 0603 (1608 Metric) - Thin Film	R6			1							
	120Ω	Description	R9			1							
	RC0402FR-0740K7L		R10, R11	CMP-3211-00586-2	Up to date	2	Bourns	CRD402-FX-4701GLF	Volume Production	Mouser	652-CRD402FX-4701GLF	0.008	0.08
	5.1Ω	Res Thick Film 0603 5.1K Ohm 1% 0.1W(1/10W) Â±100ppm/°C Epoxy Pad SMD T/R	R12, R13			2							
	RC1206JR-0722RL	22 Ohms Â±5% 0.25W, 1/4W Chip Resistor 1206 (3216 Metric) Moisture Resistant Thick Film	R14, R15			2							
	BOOT	Tactile Switch SPST-NO Top Actuated Surface Mount	S1			1							
	RESET	Tactile Switch SPST-NO Top Actuated Surface Mount	S2			1							
	TPS54821RHLR	4.5-Vin 17-V, 8-A, synchronous buck converter with Accuip current limit 14-VQFN -40 to 125	U1			1							
	AP2112K-3.3TRG1	AP2112 Series 0.6 A 3.3 V Fixed Output SMT LDO Linear Regulator - SOT23-5	U2	CMP-019-00049-1	Up to date	1	Diodes Inc.	AP2112K-3.3TRG1	Volume Production	Mouser	621-AP2112K-3.3TRG1	0.26	0.26
	LSM6DSO32TR	Accelerometer and 3D Gyroscope 3D Always-On 1.71V to 3.6V0.55mA14-Pin LGA TR	U3	CMP-010-000224-1	Up to date	1							
	ESP32-S3-WROOM-1-N16R8	Bluetooth, WiFi 802.11b/g/n, Bluetooth v5.0 Transceiver Module2.4GHz PCB Trace Surface Mount	U5			1							